

Turkish Version

Corporate Artificial Intelligence Governance in the Healthcare Sector

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Introduction

In the early 2020s, artificial intelligence (AI) in healthcare was limited to “predictive” models; today, it has evolved into a “*generative and multimodal*” structure capable of diagnosing, planning treatments, and communicating with patients in natural language. Artificial intelligence (AI) has become an integral part of clinical processes and operational management in the healthcare sector. However, this technological leap also necessitates careful consideration of serious risks. Misdiagnoses, algorithmic biases, and data breaches make an institutional governance framework imperative.

By 2026, the full implementation of regulations such as the EU AI Act will make “*Enterprise AI Governance*” a legal requirement rather than an option for healthcare organizations. *Corporate governance* refers to decision-making and oversight mechanisms that balance the interests of shareholders, stakeholders, and the public good (Millstein, 1998:27; Witherell, 2002:18). *AI governance*, on the other hand, is the set of policies, responsibilities, and oversight mechanisms that a healthcare organization employs while developing, integrating, and monitoring AI technologies.

Corporate AI Governance in Healthcare refers to the holistic management of data analytics systems, machine learning models, sensor-based hygiene infrastructure, and robotic/intensive care AI applications used in hospitals and healthcare institutions—not only in terms of technical performance but also in their strategic, legal, and ethical dimensions. It is addressed as a strategic management issue centered on corporate responsibility, accountability, and patient safety (Mantymaki, Minkkinen, Birkstedt, et al., 2022: 604–605).

AI-based systems continue to improve healthcare delivery each year in areas such as clinical decision support, image analysis, intensive care predictive models, and operational optimization. However, these systems are classified as high-risk and pose multidimensional risks related to patient safety, algorithmic bias, data privacy, and accountability (WHO, 2021; EU, 2024). For example, the 2024 EU Artificial Intelligence Act (EU AI Act) classifies diagnostic and decision-support systems used in healthcare as “*high-risk AI systems*.” This classification requires that systems undergo a comprehensive risk-benefit analysis before being made available for clinical use. In this way, the potential for harm to patients is minimized as early as the design phase.

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The widespread adoption of artificial intelligence (AI) applications in the healthcare sector necessitates not only technical risk management but also a multi-layered governance architecture at the organizational level. This study addresses the use of AI in healthcare organizations within an integrated organizational governance framework that spans from board-level oversight to clinical practice. The study follows a three-phase approach: In the first phase, the theoretical framework for corporate AI governance in healthcare is discussed based on international regulatory frameworks ; in the second phase, global implementation examples are examined; and in the final phase, a proposal for a model applicable to Turkey is developed.

1. Institutional AI Governance Structure in Healthcare Based on Existing International Frameworks

Effective corporate AI governance in the healthcare sector requires not only technical control mechanisms but also a holistic corporate structure based on layers of strategic oversight, risk management, ethical oversight, and operational accountability. At the international level, the World Health Organization (WHO, 2021), the EU AI Act (2024), ISO/IEC 23894:2023 Information Technology—Artificial Intelligence Risk Management (2023), the OECD Principles of Corporate Governance (2015), and the National Institute of Standards and Technology (NIST) AI Risk Management Framework (2023) form the normative foundation of this structure. When considered together, these documents suggest that the governance architecture to be established in healthcare organizations can be structured across four main levels. These four main levels are as follows:

1.1. Board-Level Oversight

The specified documents do not include a fully integrated, strategic-level board model for healthcare organizations. It should be noted that there is a gap in the literature in this area. There is an ISO standard for risk management in the healthcare sector, which recommends managing AI risks throughout the entire lifecycle. In the World Health Organization's (2021) document, *"Ethics and Governance of AI for Health,"* the concepts of human autonomy, transparency, accountability, and sustainability are identified as ethical principles. It does not address how a strategic board structure should be designed. The OECD Principles of Corporate Governance (2015: 44–46) state that boards of directors bear the responsibility for active oversight of strategic risks. The EU AI Act (2024), under Article 9 (*Risk Management System*), imposes high-level accountability and compliance obligations for high-risk systems; this refers to the board level in healthcare institutions. As a regulation, the EU AI Act covers the risk management system, technical documentation, human oversight, and market monitoring. It does not include a strategic corporate model. The requirement to establish a risk management system and a quality management system can be interpreted as a condition of corporate responsibility.

Since artificial intelligence applications in healthcare pose high risks in terms of patient safety, legal liability, and institutional reputation, the board of directors must assume a clear oversight role regarding the use of artificial intelligence. Its responsibilities include approving the institutional artificial intelligence strategy, identifying risk factors, monitoring high-risk clinical artificial intelligence applications, and preparing periodic *"artificial intelligence risk reports."* The AI risk report is a strategic risk assessment document intended for board oversight that comprehensively presents performance, deviation, bias, compliance, and clinical safety metrics of the organization's AI systems. It includes a list of AI systems actively used within the

institution, the system type (diagnostic support, image analysis, etc.), risk levels classified as low, medium, or high in accordance with the EU AI Act, and their distribution by clinical unit (Emergency Department, Intensive Care Unit, Radiology, etc.). In short, the board of directors bears the ultimate responsibility and compliance obligations for high-risk AI systems.

1.2 Corporate AI Committee and Organizational Structure

World Health Organization’s “Ethics and Governance of Artificial Intelligence for Health” document (2021:34–35) and the National Institute of Standards and Technology () Artificial Intelligence Risk Management Framework (2023: 13–14) recommend that corporate roles and accountability mechanisms in AI systems be clearly defined; this requirement necessitates the establishment of a structured AI governance organization within healthcare institutions. The English term “govern” used in both documents requires that organizational roles be clearly defined. In this context, the organizational structures to be established in healthcare institutions are as follows:

- Institutional Artificial Intelligence Committee (e.g., a committee comprising the Chief Medical Officer, a clinical representative, the Director of Information Technology, and the legal counsel)
- Clinical Ethics Committee
- Data Protection and Cybersecurity Unit
- Internal Audit Mechanism and Quality Committee.

The ISO/IEC 23894 (2023) standard emphasizes the need to integrate AI risk management at the organizational level. Therefore, AI governance should be integrated into the corporate risk management system rather than managed by a separate technical team.

The Corporate AI Committee is conceived as a governance body operating under the oversight of the board of directors and integrating the dimensions of strategy, risk, ethics, and technical implementation. For high-risk units such as Intensive Care and the Emergency Department, there should also be subcommittees. Its strategic responsibilities should include developing and continuously updating the organization’s AI strategy; determining AI application areas based on clinical priorities and risk levels; conducting strategic impact and risk analyses for AI investments; presenting periodic AI risk reports to the board of directors; and approving corporate AI policies and procedures.

With regard to clinical safety and validation, the institutional AI committee is responsible for approving the clinical validation process for new AI systems, verifying that a “human-in-the-loop” decision-making and oversight mechanism is in place, monitoring indicators of clinical errors, AI system bias, and model drift, making decisions regarding model recall (algorithm recall), and evaluating adverse event analyses.

If we examine these concepts individually, a *Clinical Error* is an incorrect clinical decision generated by an AI-supported system that has the potential to harm the patient. This error may stem directly from the model or from a misinterpretation of the model. For example, in an Intensive Care Sepsis model, if the model classifies a patient as “low” and the patient goes into septic shock within six hours, this constitutes a high-impact clinical error. *Model Drift* refers to changes in data or relationships that cause a machine learning model’s performance to decline

over time. Examples include minor data changes leading to false negatives and, ultimately, an increase in mortality, or measurement differences in new devices resulting in misclassification. A *model recall decision* involves suspending the use of a model when a specific performance decline or safety risk is detected. For example, if the accuracy rate for mortality prediction in an intensive care unit decreases and the false-negative rate increases, the model may be temporarily deactivated by committee decision and the system switched to manual clinical decision-making mode. An *adverse event* is a clinical event associated with an AI system that has the potential to cause harm or has caused harm. This may result directly from a model error or from a process influenced by the model. For example, a medication dosing algorithm incorrectly assesses renal function. This leads to a recommendation for an excessive dose. As a result, the patient experiences acute kidney injury. The algorithm's incorrect assessment leads to clinical errors. Therefore, Article 72 of the EU AI Act (2024) and Clauses 6 and 8 of ISO 23894 (ISO/IEC 23894, 2023) mandate the establishment of continuous risk management and monitoring mechanisms; this requirement necessitates structured organizational governance systems at the institutional level.

In the context of data and privacy management, the committee is expected to monitor compliance with the Personal Data Protection Law (KVKK), implement data anonymization and security procedures, evaluate privacy-preserving techniques such as federated learning, and initiate the crisis management process in the event of a data breach (GDPR, 2016/679; NIST, 2023). Here, *federated learning* is a distributed machine learning method developed specifically to address data privacy in the healthcare sector, which enables a model to be trained without data being transferred to a central server—instead, the data remains at the institutions where it is stored. For example, the Ministry of Health wishes to develop a sepsis model across 20 public hospitals. It proposes an initial model. The 20 hospitals train the model using their own intensive care data. Without patient data leaving the hospital, the hospitals only send model parameter updates. This is federated learning. From a data perspective, the principle of *Data Minimization*—which requires that personal data be collected and processed only to the extent necessary for a specific, explicit, and legitimate purpose—is crucial here. Article 5(1)(c) of the EU General Data Protection Regulation (GDPR, 2018) states that “data *must be adequate, relevant, and limited to what is necessary for the purpose,*” and Article 4 of the Turkish Data Protection Law (KVKK, 2016) stipulates that “data *must be relevant to the purpose for which it is processed, limited, and proportionate.*” When collecting data for a patient in a healthcare setting, since social media and address data are not necessary, in the case of the sepsis model, for example, only pulse, blood pressure, lactate, and temperature are processed if these are sufficient.

As part of the committee's oversight of ethics and fairness, its responsibilities include reviewing analyses of algorithmic bias, coordinating with the ethics committee, evaluating demographic performance disparities, and monitoring the implementation of the principles of transparency and explainability. *Algorithmic bias* occurs when an artificial intelligence model systematically performs better or worse for a specific group (e.g., age, gender, socioeconomic status). If the model does not work equally for everyone, bias exists. For example, if a sepsis model has an accuracy of 90% for male patients but only 82% for female patients, there is a clinical bias. In this case, the committee is expected to measure model performance by demographic subgroups, compare error rates, and conduct a technical and ethical assessment if there are significant differences. The primary goal of ethical and equity oversight is to determine whether an AI model has the potential to harm a specific group. It is crucial to ensure *distributive justice*,

whether healthcare is distributed equitably; *procedural justice*—that the decision-making process is transparent; and *clinical justice*—that false negatives are not concentrated in a specific group. Regarding algorithmic bias, the World Health Organization’s (2021) “*Ethics and Governance for Health*” document, on page 27, emphasizes that AI systems carry the risk of reproducing existing health inequities, that a lack of representation in datasets can lead to discrimination, and that justice must be upheld for vulnerable groups. On page 34 of the same document, it is stipulated that institutions have an obligation to monitor the risks of bias in AI systems and that oversight mechanisms must be established for this purpose. Similarly, the 2024 EU Artificial Intelligence Act mandates the monitoring of discrimination risks in high-risk systems. The Corporate AI Committee is therefore responsible for assessing discrimination risks as part of its ethics and fairness oversight.

1.3. Risk and Compliance Management Layer

Corporate AI governance in healthcare must be aligned with regulatory and risk frameworks. In this context, in accordance with ISO 31000 and ISO/IEC 23894, risks must be identified, analyzed, continuously monitored, and risk mitigation plans implemented (SG Systems, 2025). Within the risk cycle, the stages of context, identification, analysis, assessment, mitigation, and monitoring are applied to AI systems. Attention is drawn to the importance of determining what deviations, biases, hallucinations, “black box” decisions, and malicious inputs stemming from AI might entail, and how to control them.

In addition to risk management, regulatory compliance must also be ensured. Regulatory compliance is not merely a legal requirement but a strategic safeguard mechanism for patient safety. Compliance encompasses the Personal Data Protection Law (KVKK) and the EU General Data Protection Regulation (GDPR) in the field of data protection; the EU Artificial Intelligence Act for high-risk system requirements; the U.S. Food and Drug Administration (FDA) SaMD (Software as a Medical Device) guidelines for medical software; and medical device regulations under the Medical Device Regulation (MDR). Regarding SaMD, examples include software that enables mobile devices to display MRI, ultrasound, or X-ray images used for diagnostic purposes; software that processes images to help detect breast cancer; and software that collects patient data in real time, is monitored by a healthcare professional, and is used to develop treatment plans.

It is recommended that the risk and compliance layer in healthcare institutions be managed not by a single individual, but by a multi-tiered organizational structure. Strategic responsibility lies with the hospital’s board of directors. Operational execution may be assigned to the institutional artificial intelligence management committee. The data protection officer oversees data processing activities under the GDPR/KVKK, reports data breaches, and ensures compliance with the principle of data minimization. The quality management unit may also be involved in ensuring regulatory and medical device compliance.

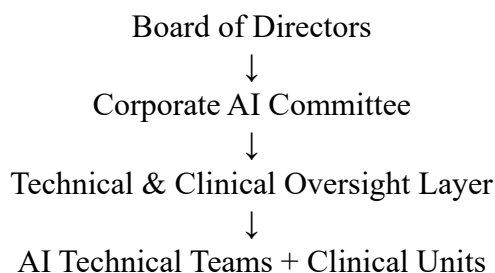
1.4. Technical and Clinical Oversight Layer

The final layer, the technical and clinical oversight layer, is an operational oversight structure that reports to the Corporate AI Committee and monitors both the technical reliability and clinical safety of AI systems. This layer is not a decision-maker; rather, it serves as a control mechanism that analyzes, verifies, and reports. The technical oversight sub-layer may consist of data scientists, machine learning engineers, AI systems engineers, cybersecurity experts, quality management specialists with knowledge of SaMD and MDR, and a data protection representative.

The technical audit group’s responsibilities include AI model performance monitoring, model drift monitoring, bias analysis, technical security, and documentation creation. Model performance monitoring involves accuracy, confidence interval analysis, performance threshold tracking, and measurements used in AI performance assessment in healthcare, such as AUC (Area Under the ROC Curve), FNR (False Negative Rate), and FPR (False Positive Rate). FNR indicates the proportion of people who are sick but are classified as “healthy” by the model. FPR, on the other hand, indicates the proportion of people who are actually healthy but are classified as “sick” by the model. In intensive care, a low FNR is expected, while in the emergency department, a balanced FNR/FPR ratio is expected. As previously mentioned, *model drift monitoring* refers to the performance degradation that occurs when the difference between the data distribution used to train the model and the new data distribution encountered in the real-world increases. As part of this monitoring, data change analysis and are used to check whether the drift threshold has been exceeded. In the technical security section, cybersecurity vulnerability analysis and unauthorized access tracking are performed.

The clinical oversight subgroup is expected to include the chief medical officer, relevant specialty physicians, the clinical quality officer, and a clinical researcher. Their primary responsibilities include the clinical interpretation of AI model outputs, analysis of false negatives, and clinical validation—including the assessment of mortality and morbidity effects; ensuring that physicians remain the final decision-makers; analyzing potential AI-related harm; determining clinical risk thresholds; and providing training to ensure physicians use the model correctly. In short, the Technical and Clinical Oversight Layer is an operational oversight mechanism composed of interdisciplinary experts that continuously monitors the performance, safety, ethical, and clinical impacts of AI systems in healthcare institutions.

Based on the documentation, organizational AI governance establishes the following management structure:



2. New Job Roles in Corporate AI Governance

The full implementation of enterprise AI governance necessitates the definition of new roles and responsibilities that require expertise beyond traditional hospital organizational structures. Traditional hospital structures are organized around clinical service delivery, administrative management, quality control, and information technology functions. However, AI systems involve multi-layered risk areas such as technical algorithmic risk, clinical safety, data protection, ethical responsibility, and regulatory compliance. For example, issues such as model drift, algorithmic bias, or autonomous decision-support mechanisms fall within the purview of neither the IT department alone nor the clinical quality team alone. Consequently, there is a need for hybrid expertise—combining interdisciplinary, technical, and clinical knowledge—to monitor and manage AI-specific risks.

In addition, the EU AI Act, ISO/IEC 23894, and the World Health Organization’s (2021: 24–26) principles for AI governance in healthcare require risk management, continuous monitoring, and accountability mechanisms throughout the entire lifecycle. These requirements necessitate institutional-level ownership of functions such as “AI risk responsibility,” “algorithmic fairness oversight,” or “model performance monitoring”—roles that have traditionally been poorly defined in hospitals.

The “human-in-the-loop” requirement limits the ability of AI in healthcare to fully replace humans in an autonomous manner, both from a legal and ethical perspective. Specifically for high-risk systems, the EU AI Act (2024, Article 14) explicitly mandates human oversight and stipulates that the ultimate responsibility for decisions must remain with a qualified human operator. Since healthcare is directly related to patient safety, the interpretation of algorithmic decisions within a clinical context, the evaluation of exceptional cases, and the assumption of responsibility can only be carried out by a physician or an authorized healthcare professional. For this reason, AI systems are positioned as “*decision support* systems” in healthcare and do not replace the final clinical decision.

Furthermore, healthcare is not merely a process of data analysis; it requires ethical judgment, empathy, consideration of patient preferences, and clinical intuition in the face of uncertainty. While artificial intelligence can identify statistical patterns with high accuracy, it cannot, on its own, assess individual patient contexts, sociocultural factors, or value-based dimensions of decision-making. The World Health Organization (2021) also emphasizes the protection of human autonomy as a fundamental principle in the use of AI in healthcare. Within this framework, the “human-in-the-loop” approach envisions not the replacement of healthcare professionals by AI, but rather the creation of a hybrid clinical model that enhances their decision-making capacity and maintains responsibility centered on the human.

The new job positions created by organizational AI governance in the healthcare sector, along with their job descriptions, are as follows in the table below (Fountaine, McCarthy & Saleh, 2019; Raisch & Krakowski, 2021);

Position	Why Is It New? (What Is Missing in the Traditional Hospital Structure)	Core Duties / Responsibilities	Governance Layer
AI Quality and Validation Specialist	Traditional quality units audit medical devices or processes; they lack expertise in algorithmic model validation.	Clinical validation design, model performance testing (AUC, FNR), MDR/SaMD compliance checks, version verification	Technical & Clinical Audit
Health Data Governance Officer	Artificial intelligence data architecture, data standardization, and data quality each require specialized expertise.	Data minimization, data quality control, federated learning coordination	Data & Compliance
Artificial Intelligence Incident Response Officer	Hospitals have clinical incident management systems, but there is no defined process for managing AI-related technical-clinical hybrid incidents.	Analysis of AI-related adverse events, model suspension process, crisis escalation, root cause analysis (RCA)	Risk & Security
Explainable Artificial Intelligence (XAI) Specialist	IT teams develop models, but expertise in clinical explainability and transparency is generally lacking.	Clinical explanation interface, algorithmic transparency report, human decision support	Ethics & Clinical

Position	Why Is It New? (What Is Missing in the Traditional Hospital Structure)	Core Duties / Responsibilities	Governance Layer
AI Vendor Governance Manager	Hospitals purchase software but lack expertise in algorithmic risk and data contract oversight.	AI contract clauses, third-party risk analysis, model update obligations, data processing agreements	Compliance & Strategy
AI Risk and Compliance Manager	Existing risk units do not systematically monitor AI lifecycle risks.	EU AI Act compliance, ISO 23894 integration, risk matrix, coordination of model recall decisions	Enterprise Risk
AI Operations Manager	Model development exists, but there is no separate role for monitoring the live environment and managing drift.	Drift monitoring, performance metrics, continuous model updates	Technical Operations
Clinical AI Safety Officer	There is a patient safety unit, but expertise in AI-specific clinical risk analysis is lacking.	False-negative analysis, mortality impact assessment, tracking of clinical readmission rates	Clinical Safety

Consequently, the new positions listed in the table are not the result of technological expansion; they are the institutional response to the multi-layered risk structure of AI systems. While traditional organizational charts operate within structures where clinical services, information technology, and legal compliance functions are separate silos (a structure where units within an organization operate independently of one another, with limited communication and without holistic coordination), while AI systems are positioned at the intersection of these areas and require the simultaneous management of technical performance, data governance, ethical oversight, patient safety, and regulatory compliance. Consequently, hybrid roles such as the AI Quality and Validation Specialist, Health Data Governance Officer, or AI Incident Response Officer have emerged as a structural necessity to strengthen organizational accountability, monitor risks throughout their lifecycle, and effectively implement a human-in-the-loop approach. Consequently, these positions serve as the organizational guarantee for the sustainable, secure, and regulatory-compliant operation of corporate AI governance.

3. Examples of Corporate AI Governance in the Healthcare Sector Around the World

Although corporate AI governance in healthcare has not yet reached a uniform and standardized model on a global scale, practices developed in various countries and leading healthcare institutions demonstrate that risk-based, multilayered governance structures centered on human oversight are becoming increasingly institutionalized. Below is a summary of some notable national examples of corporate AI governance highlighted in international literature.

U.S. Practices

a) Mayo Clinic Artificial Intelligence Governance Framework (Rochester, Minnesota)

The Mayo Clinic has developed an institutional artificial intelligence governance framework to systematically evaluate clinical AI applications. The *Artificial Intelligence* Review Board, established within the organization, subjects AI projects to a multi-tiered evaluation based on criteria such as technical accuracy, clinical validity, ethical appropriateness, and patient safety.

A validation process is conducted before the AI is put into clinical use; after implementation, performance, model drift, and safety metrics are continuously monitored. A “human-in-the-loop” approach is required, and algorithms are positioned as decision support systems (Halamka & Chandra, 2021: 1241–1242).

b) Harvard–MIT Artificial Intelligence Oversight Model (Boston–Cambridge, Massachusetts)

The Harvard Medical School and MIT have jointly adopted the Artificial Intelligence Oversight Model (Clinical AI Oversight Model). This model involves a committee process that ensures artificial intelligence systems developed or used in clinical settings are systematically evaluated in terms of technical accuracy, clinical validity, risk of bias, ethical appropriateness, and workflow integration. It mandates that algorithms undergo separate risk analyses during the development, validation, and deployment phases. A distinctive feature of the model is that it evaluates not only technical performance but also integration into clinical workflows and potential bias risks. The integration of the ethics committee and clinical leadership into the process ensures that governance is conducted within an interdisciplinary framework (Li, Reis, Rodman, et al., 2026).

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c) Stanford – Clinical AI Review Process (California)

Stanford University Medical Center has established a structured review process for clinical artificial intelligence projects. In this process, AI applications are tested through pilot studies and their performance is validated using real patient data before being deployed for clinical use. Post-deployment monitoring mechanisms ensure that model performance and potential risks are regularly reviewed. The Stanford model offers a balanced approach between innovation and institutional risk management (Handler, 2026; Brodeur, Goh, Rodman, and Chen, 2026).

United Kingdom

United Kingdom National Health Service (NHS) – Artificial Intelligence Assurance Framework

The United Kingdom’s National Health Service (NHS) has developed a standardized assessment structure for the use of artificial intelligence in healthcare through its Artificial Intelligence Assurance Framework (AI Assurance Framework). This framework provides comprehensive checklists covering safety, data protection, ethical compliance, transparency, and performance criteria. A risk-based approach is adopted, and stricter clinical validation and human oversight are required for high-risk systems. While national-level guidance is provided, implementation is left to local hospital committees (Gov UK, 2024).

Germany

MDR-Compliant Clinical AI Governance

In Germany, the governance of artificial intelligence in healthcare is shaped within the framework of the European Union’s Medical Device Regulation (MDR). In particular, AI systems developed in collaboration between university hospitals and technology companies are classified as medical devices and are therefore subject to post-market surveillance processes that require rigorous technical documentation, clinical validation, and continuous monitoring of performance after the product is placed on the market. The German approach offers a robust,

quality-based governance model that integrates regulatory compliance with technical reliability (Grieb, Woitz, Farber, and Lebrecht, 2026).

China

Smart Hospital and Centralized Digital Health Model

In China, AI governance is largely carried out within the framework of state-supported digital health strategies. Under “*Smart Hospital*” programs, clinical decision support and image analysis systems are monitored and updated through centralized platforms. The key feature of this model is a nationally integrated data infrastructure and a centralized coordination mechanism. Governance is primarily state-centered, providing advantages in terms of scale and rapid digitalization.

One of the most notable examples of AI applications in healthcare in China is the *Ping An Good Doctor platform* and the “*One-Minute Clinics*” model developed in conjunction with it. These small kiosks, which are becoming widespread across China, represent a marvel of AI-based healthcare governance. Developed by Ping A Healthcare Technology, this system provides AI-assisted preliminary diagnosis and digital health counseling services through physical kiosks and the mobile platform. Known as “One-Minute Clinics,” these smart health stations allow patients to enter their symptoms via a digital interface, conduct preliminary assessments using AI-based triage algorithms (a rule- or model-based decision system that classifies patients according to the urgency of their condition in healthcare settings), and refer patients to a remote physician consultation when necessary. This system aims to increase rapid access to healthcare services, particularly in rural areas and densely populated urban centers, while also providing services such as basic health counseling and over-the-counter medication procurement in a digital environment (He et al., 2019, p. 653; Ping A Healthcare Technology, 2020).

The Ping A Good Doctor platform also enables the continuous monitoring and updating of artificial intelligence systems through its extensive data infrastructure and centralized algorithm management. The platform develops machine learning models using anonymized health data collected from millions of users and optimizes clinical decision-support processes. In this regard, the system is viewed not only as a digital healthcare service delivery platform but also as a healthcare ecosystem based on large-scale data analytics. Developed in alignment with China’s “*Smart Hospital*” strategy, this model aims to enhance the accessibility and efficiency of healthcare services by integrating AI-supported triage, remote consultation, and data-driven healthcare management applications (Wang, 2025).

4. Proposed Model for a Framework of Institutional AI Governance in Healthcare for Turkey

The fact that healthcare services in Turkey are largely administered within a centralized public management structure and that a robust digital health infrastructure (e-Nabız, MHRS, centralized hospital information management systems) is in place presents a unique opportunity for institutional artificial intelligence (AI) governance. However, given the patient safety, data protection, and legal liability dimensions of high-risk AI applications, there is a need for a structured and multi-layered governance model. Within the scope of this study, a risk-based, human-centered, and lifecycle-focused institutional AI governance model can be proposed for Turkey.

The proposed model is a multi-layered governance model consisting of two main levels: (i) the national policy and standard-setting level, and (ii) the hospital-level operational governance level. *At the national level*, it is recommended that a “*National Health Artificial Intelligence Coordination Council*” be established within the Ministry of Health. The primary responsibilities of this council include determining risk classification criteria for AI systems, defining technical validation standards, preparing data governance guidelines, and establishing minimum safety thresholds for high-risk applications. This structure should be designed in accordance with the risk-based approach of the EU Artificial Intelligence Regulation and the life cycle risk management principles of ISO/IEC 23894 (ISO/IEC 23894, 2023, Clauses 6 and 8).

At the hospital level, it is recommended that an “*Institutional Artificial Intelligence Committee*” be established under the Chief Medical Officer. This committee should consist of a representative of the Chief Medical Officer, an IT manager, a legal advisor, a data protection officer (under the scope of the Personal Data Protection Law), a representative of the relevant clinical unit, and a quality management specialist. The committee should be responsible for conducting preliminary assessments of AI projects, coordinating the clinical validation process, monitoring model performance, and preparing periodic AI risk reports.

Life-cycle-based risk management requires that AI systems be monitored not only during the development phase but also throughout their operational lifespan. A three-stage assessment process may be appropriate for an AI model (EU, 2024, Articles 9, 14, 72). These are:

1) Preliminary Assessment and Validation:

The model’s technical performance (e.g., AUC, FNR, FPR), data representativeness, and potential algorithmic bias are analyzed. A pilot study is required prior to clinical integration.

2) Deployment and Clinical Integration:

The AI system is used solely for decision support, and a human-in-the-loop mechanism is required (WHO, 2021: 25, 34). The clinical override rate (the human decision-making process that overrides the AI decision) and its impact on workflow are monitored.

3) Continuous Monitoring and Updates:

Model drift (performance decline), bias indicators, and adverse events are reported regularly. If predefined threshold values are exceeded, the model is temporarily paused or retrained.

Regarding data management and legal compliance, since health data is defined as special-category personal data under Article 4 of Turkey’s Personal Data Protection Law No. 6698 (KVKK), the principles of data minimization, explicit consent, and data security must be central to the model. In this context, it is recommended to evaluate privacy-preserving techniques such as federated learning and limit centralized data transfers. The integration of data governance processes with technical performance monitoring mechanisms is of critical importance for strengthening institutional accountability (EU, 2016, Article 5).

In the proposed model, AI systems are positioned as tools that do not replace physician decision-making but rather support decision-making capacity. In line with the “human-in-the-loop” approach, it must be mandatory for an authorized healthcare professional to bear ultimate decision-making responsibility in every high-risk AI application. It is recommended that

clinical override rates be monitored regularly and that unusual increases be evaluated as risk indicators.

In conclusion, this model offers a hybrid governance approach that combines centralized policymaking with hospital-based committee oversight. Through a structure that integrates risk-based classification, lifecycle monitoring, and human oversight, the aim is to strengthen both patient safety and legal compliance. This will enable the development of AI applications in healthcare in Turkey within a systematic, transparent, and sustainable institutional framework.

Conclusion

The rapid proliferation of AI applications in the healthcare sector necessitates that these technologies be addressed not merely as technical innovations, but also as a strategic area requiring management at the institutional level. Applications such as diagnostic support systems, image analysis, clinical decision support tools, and telehealth services have the potential to increase the efficiency and accessibility of healthcare services; however, they also bring with them significant risk areas such as patient safety, data protection, ethical responsibility, and legal accountability. Therefore, developing an organizational AI governance approach has become a critical requirement for the sustainable and safe implementation of AI in healthcare institutions.

Within the scope of this study, the core components of institutional AI governance in healthcare—strategic governance, data governance, model governance, and the ethical-legal framework—are addressed within a multi-layered structure. The strategic governance dimension is important in terms of institutions defining their vision for AI use and identifying risk-based application areas. Data governance involves the secure processing and anonymization of health data, as well as ensuring data quality; model governance, on the other hand, encompasses the clinical validation of AI systems, performance monitoring processes, and the management of technical risks such as model drift. The ethical and legal framework includes principles such as algorithmic transparency, shared responsibility, and human oversight.

An examination of international practices reveals that AI governance in healthcare has evolved through various institutional models. Committee-based oversight models developed by academic medical centers in the U.S., assurance frameworks established at the national level in the United Kingdom, and a regulation-based approach in Europe highlight the diverse governance approaches in this field. In China, meanwhile, an AI ecosystem based on large-scale data infrastructure is being developed through smart hospital strategies and digital health platforms. These examples demonstrate that AI governance in healthcare cannot be reduced to a single model, but that common principles—such as a risk-based approach, continuous monitoring, and human oversight—are paramount.

From Turkey's perspective, its robust digital health infrastructure and centralized health management system offer significant potential for institutional AI governance. However, to ensure that AI applications can be safely integrated into healthcare services, it is necessary to establish internal governance structures, develop technical and ethical oversight mechanisms, and institutionalize lifecycle-based risk management processes. Holistic governance models developed in this direction will contribute to the safe, transparent, and sustainable integration of AI technologies into the healthcare system.

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